

Euclid, Book I, Proposition I.6

If Two Angles of a Triangle Are Equal, Then the Opposite Sides Are Equal

CGJteamLab

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1 Introduction

Proposition I.6 establishes the converse of the preceding proposition. Where Proposition I.5 proves that equal sides of a triangle imply equal base angles, Proposition I.6 shows that equal base angles imply the equality of the opposite sides.

This result completes the fundamental correspondence between the sides and angles of an isosceles triangle. Together, Propositions I.5 and I.6 establish the equivalence between the two defining properties of an isosceles triangle.

Euclid's original proof is considerably more involved than the modern reconstruction. It proceeds by contradiction, introducing an auxiliary construction obtained from Proposition I.3 and ultimately deriving an impossible configuration by repeated use of Proposition I.4.

In the Hilbert reconstruction developed in this project, the theorem admits a substantially shorter proof. The previously established Angle–Side–Angle theorem is sufficient to derive the equality of the opposite sides directly, eliminating the auxiliary construction and the *reductio ad absurdum* argument required by Euclid.

2 The Classical Construction

Euclid's proof begins with a triangle whose base angles are assumed to be equal. The desired conclusion is that the opposite sides are also equal.

Instead of arguing directly, Euclid assumes that the two sides are unequal and derives a contradiction. The proof therefore combines an auxiliary construction with a *reductio ad absurdum* argument.

The following sequence illustrates the successive stages of Euclid's construction.

2.1 Initial Triangle

Let ABC be a triangle satisfying

$$\angle ABC \cong \angle ACB.$$

At this stage no auxiliary construction has yet been introduced.

2.2 Assuming Unequal Sides

To obtain a contradiction, Euclid assumes that the two sides opposite the equal angles are not equal. Without loss of generality, suppose

$$AB > AC.$$

Using Proposition I.3, a point D is chosen on the segment AB so that

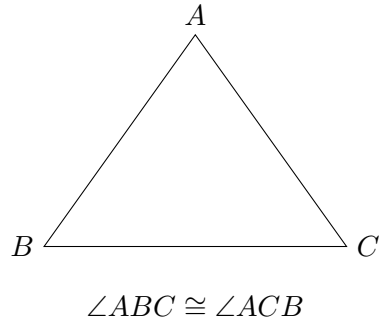


Figure 1: The initial triangle with equal base angles.

$$DB \cong AC.$$

Consequently,

$$A - D - B.$$

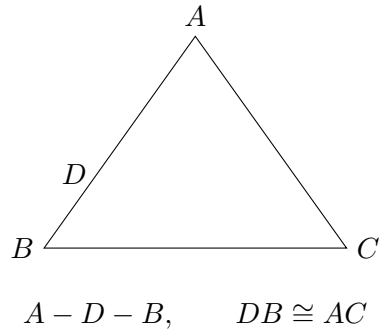


Figure 2: Assuming $AB > AC$, Proposition I.3 constructs D on AB with $DB = AC$.

2.3 The Auxiliary Triangle

The auxiliary point is joined to the remaining vertex of the original triangle.

Since

$$DB \cong AC,$$

and the angle

$$\angle DBC$$

coincides with the original angle

$$\angle ABC,$$

the auxiliary triangle is prepared for an application of Proposition I.4.

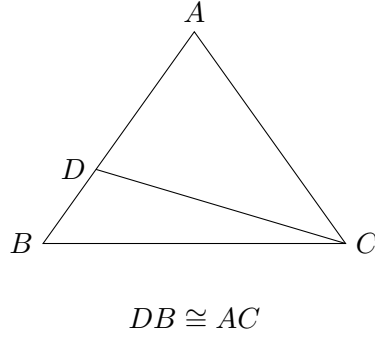


Figure 3: The auxiliary triangle DBC obtained after joining D to C .

2.4 The Contradiction

Applying Proposition I.4 to the triangles DBC and ACB establishes their congruence. Consequently, the corresponding angles and sides are equal.

Since the point D lies between A and B , one of the resulting equalities contradicts the assumption that the whole segment AB is strictly greater than its proper part DB .

The contradiction shows that the assumption

$$AB > AC$$

is impossible. By symmetry, neither side can be greater than the other, and therefore the two sides are equal.

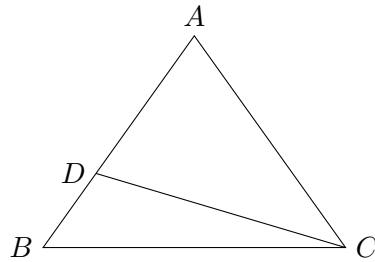


Figure 4: The auxiliary configuration from which Proposition I.4 forces a contradiction with the assumption $AB > AC$.

3 Statement

Theorem 1 (Euclid I.6). *Let ABC be a nondegenerate triangle.*

If

$$\angle ABC \cong \angle ACB,$$

then

$$AB \cong AC.$$

That is, equal angles in a triangle subtend equal opposite sides.

4 Proof

Euclid proves the theorem by contradiction.

Assume that the sides opposite the equal angles are unequal. Without loss of generality, suppose that AB is greater than AC . By Proposition I.3, a point is constructed on the longer side so that the remaining segment is equal to the shorter side.

Joining the auxiliary point to the remaining vertex produces a second triangle. Proposition I.4 is then applied to show that the auxiliary triangle is congruent to the original one.

The resulting congruence forces equalities that are incompatible with the assumption that one side is strictly longer than the other. Since the assumption leads to a contradiction, the two sides must be equal.

Thus equal angles in a triangle necessarily subtend equal opposite sides.

5 Implementation Architecture

The formal proof follows a substantially different route from Euclid’s classical argument. Rather than introducing an auxiliary point, constructing additional triangles, and reasoning by contradiction, the implementation exploits a higher-level theorem already available in the Hilbert interface.

The proof compares the given triangle with itself after exchanging the roles of the two equal angles. Since the common side is trivially congruent to itself, the derived Angle–Side–Angle theorem immediately implies the equality of the remaining sides.

Consequently, the entire classical construction disappears from the formal proof and is replaced by a single application of a previously established congruence theorem.

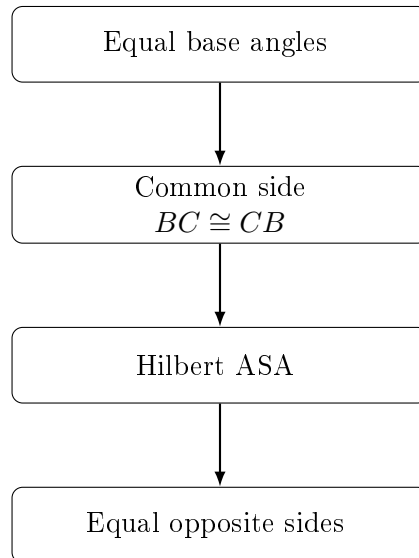


Figure 5: Architecture of the formal reconstruction of Proposition I.6.

6 Hilbert Reconstruction

The formal reconstruction replaces Euclid’s indirect argument by a direct consequence of the developed Hilbert theory.

The key observation is that the given triangle can be compared with itself after exchanging the two equal base angles. The side BC is common to both descriptions of the triangle and is therefore congruent to itself.

The hypotheses of the derived Angle–Side–Angle theorem are thus immediately satisfied:

$$\triangle BCA \longleftrightarrow \triangle CBA.$$

Applying the Hilbert ASA theorem yields the congruence of the remaining corresponding sides,

$$BA \cong CA,$$

which is equivalent to

$$AB \cong AC.$$

Unlike Euclid’s original proof, no auxiliary construction, no appeal to Proposition I.3, and no proof by contradiction are required. The entire argument is reduced to a single application of a previously established triangle congruence theorem.

7 Lean Formalization

The Lean implementation closely follows the Hilbert reconstruction.

The proof first rewrites the given triangle by exchanging the roles of its equal base angles. The common side BC is identified with itself using reflexivity of segment congruence, while standard symmetry properties convert the supplied angle congruence into the orientation required by the Angle–Side–Angle theorem.

A single application of the derived theorem `hilbert_asa_sides` then establishes the congruence of the remaining sides. Finally, elementary symmetry of segment congruence rewrites the result into the statement of Proposition I.6.

Compared with Euclid’s original argument, the formal proof is substantially shorter because it relies on previously developed interface theorems rather than reconstructing the auxiliary geometric configuration.

8 Discussion

Proposition I.6 provides one of the clearest examples of the difference between a classical geometric proof and its modern formal reconstruction.

Euclid establishes the theorem by introducing an auxiliary point, constructing an additional triangle, and arguing by contradiction. The proof occupies a central place in Book I because it demonstrates how previous propositions can be combined to derive a converse statement.

Within the Hilbert framework, the same mathematical result becomes an almost immediate consequence of the previously established Angle–Side–Angle theorem. Once ASA has been derived from Hilbert’s axioms, the converse property of isosceles triangles follows directly, without auxiliary constructions or *reductio ad absurdum*.

This proposition therefore illustrates an important principle of the CGJteamLab reconstruction: as the interface theory grows, increasingly complex classical arguments collapse into short applications of high-level synthetic theorems.